

38. Length of Longest Word

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>


int main()
{
    char sentence[300];
    char word[100];
    char longest[100];


    int i = 0, j;
    int maxLength = 0;


    printf("Enter sentence: ");
    fgets(sentence, sizeof(sentence), stdin);


    while (sentence[i] != '\0')
    {
        if (isalpha(sentence[i]))
        {
            j = 0;


            while (isalpha(sentence[i]))
                word[j++] = sentence[i++];


            word[j] = '\0';
```

```

        if (strlen(word) > maxLength)
        {
            maxLength = strlen(word);
            strcpy(longest, word);
        }
    }
    else
    {
        i++;
    }
}

printf("\nLongest word = %s\n", longest);
printf("Length = %d\n", maxLength);

return 0;
}

```

Output

```

Enter sentence:  Compiler Design Laboratory

Longest word = Laboratory
Length = 10

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Process exited after 11.31 seconds with return value 0
Press any key to continue . . . |

```